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Artificial Intelligence and Machine Learning Based ChatBot

CSE 299
Junior Design Project

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DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

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Abstract—This project presents the development of a Python-based AI chatbot designed to function as an intelligent conversational assistant, capable of learning from user interactions. The chatbot integrates a local JSON-based knowledge base, structured training file, and optional web scraping functionality to deliver context-aware responses. Core natural language processing (NLP) techniques, such as TF-IDF and cosine similarity, are used to match user queries with stored data, allowing for the flexible handling of diverse phrasing.

Enhancements include mathematical problem-solving using SymPy, multilingual support with Bengali and Hindi language models, and the integration of SpaCy for improved language understanding. Advanced features, such as rental price prediction for Dhaka properties, English text summarization using a distilled BART model, and medicine database retrieval using vector embeddings and Faiss search, further expand its capabilities. A modular command-line interface enables the direct invocation of specific functionalities, thereby enhancing usability.

The chatbot's architecture emphasizes maintainability through modular code structure and robust dependency management via virtual environments. This comprehensive and scalable design enables real-world applicability across domains, laying a strong foundation for future development in intelligent and versatile chatbot systems.

Index Terms—Artificial Intelligence, Machine Learning, Chatbot, Natural Language Processing, Multilingual Support

I. INTRODUCTION

A. Background and Motivation

Conversational AI has emerged as one of the central elements of human-computer interaction in recent years and has drastically changed such areas of activity as customer service, education, and healthcare. As conversational agents, chatbots imply a natural language processing (NLP), and machine learning (ML) to simulate an artificial human-like dialogue. Their usefulness is especially relevant towards the automation of information and retrieval, domain-specific operations and the availability of 24/7 service with no human involvement.

The reason to implement this project is the necessity of smart and application-adjusted chatbot frameworks that would give effective, real-time aid over a variety of errands. Current chatbots are not very flexible, not multilingual, and/or do not have the capability of doing any context-sensitive tasks like mathematical calculation, language translation, and retrieving structured data. The project would hence close that gap by

creating a chat-bot which can easily be trained and extended as time goes on.

Besides, the increasingly fast pacing of regional and world-wide demands in multilingual societies, particularly in such nations as Bangladesh and India, precondition the development of mechanisms capable of comprehending such languages as Bengali and Hindi. Also, the use of machine learning models related to specific tasks, e.g. rent prediction or search of medicine information, demonstrates an increasing amount of so-called conversational agents that are capable of solving real-life, niche-related tasks.

B. Purpose and Goal of the Project

The main goal of this project would be the creation of an AI chatbot that would be intelligent and flexible in conversation using Python. Among the features that the chatbot facilitates are:

- Cumulatively indexed knowledge based interactive Q&A via NLP (TF-IDF + cosine similarity)
- Possibilities of solving mathematical inquiries through symbolic mathematics through SymPy
- Deep translation libraries in order to translate between Bengali and English
- The next step will be to forecast the price of the rented property based on machine learning in Dhaka (Random Forest)
- The summarization of English texts with the help of distilled BART model (DistilBART)
- A search with a structured medicine database and a semantic search answering medical queries
- The main outputs of this project have been:
Bengali and Hindi support including multilingual option
- Task-specific invocation modular command-line interface
- Domain intelligence with the use of rental prediction and medicine retrieval
- Lightweight deployment, that is fitting to be used locally and does not require extensive computing resources
- The capability of user-trainable architecture that is able to learn on the fly based on user actions

The innovation of the project is the fact that it has a unified scalable design which combines the natural conversation with

tailored and task execution. This chatbot is more useful in comparison with many current solutions that can be used exclusively as a fixed chat or in the single-domain interaction.

C. Organization of the Report

The rest of this report will be divided in the following manner:

Chapter 2: Literature Review - Reviews previous literature and passages in the field of AI chatbots, NLP and task-based virtual assistants.

Chapter 3: Methodology -An explication of the system architecture, tools, techniques, and models of chatbot development.

Chapter 4: Implementation- The process of implementing the chatbot is described and key elements such as the knowledge base, mathematical solver, language models, and external modules are described.

Chapter 5: Results and Evaluation- Gives the results of the testing of the programs, measurements of model performance and user feedback.

Chapter 6: Limitations and Challenges- Describes technical and practical difficulties that were met during development and presents existing limitations.

Chapter 7: Conclusion and Future Work- Concludes the project and gives possible ideas of improvements and future add-ons.

II. RESEARCH LITERATURE REVIEW

A. Existing Research and Limitations

Modern Natural Language Processing (NLP) and conversational AI have given rise to the development of many chatbot systems that can potentially be used to accomplish a variety of language-related tasks in an intelligent way. Major works in this area have enhanced the development of such systems with the use of transformer based models, large language data, and multilingualism skills.

A large-scale pre-trained conversational response generation model DialoGPT is created by **Zhang et al. (2020)**, and it is fine-tuned on 147 million conversations scraped on Reddit. The model scored well when it comes to producing well contextually related exciting dialogues. The training was mainly done using English data, thus the model is not useful in multilingual or Indian and regional languages like Bangla or Hindi.

Ramesh et al. (2022) showcased a multilingual transformer-based chatbot, based on the mBERT (Multilingual BERT) and tested it in seven languages. They demonstrated how well multilingual embeddings worked in cross-lingual comprehension. However, this system did not facilitate generative responses and the systems were greatly dependent on template-based retrieval models and did not allow flexibility of conversational aspect.

Saha and Islam (2021) presented a prototype Bangla chatbot that uses sequence-to-sequence RNN with a home-grown corpus of Bangla dialogues trained. Although the model trained on 80 percent basic conversational coherence data,

unavailability of pre-trained Bangla models and small amount of data had profound impact on model performance of real world applications. In addition, it did not do complex element run NLP tasks, e.g. summarization, translation or domain-specific information retrieval.

Kumar et al. (2023) created the domain specific health chatbot that had the ability of retrieving drug information based on entity recognition and fuzzy matching. The system was able to respond to user questions related to symptoms and medications but it could only communicate in English, and it had no natural flow because it was designed as rule-based system.

Research Limitations Identified:

After conducting the abovementioned literature review a number of limitations have come to be:

- 1) **Scope of Language:** The majority of developed chatbots are trained and tested on English-only. The advanced conversational systems lack a significant backing to support languages in South Asia such as Bangla and Hindi.
- 2) **Task-Specific Concentration:** Most of the studies are centered on one-task chatbots (e.g., only QA, translation, or dialogue) and are not involved in multi channel NLP tasks like summarization, prediction, and math-solving in the same entity.
- 3) **Use of Models:** Most models use classical NLP/traditional RNN methods kind of instead of not fully utilizing state-of-the-art transformers or large language models via APIs.
- 4) **Real-World Integration:** Very few research papers have shown how well NLP models integrate with the real world through interactive usage with the users (real-time), on a CLI interface.
- 5) **No Training Infrastructure:** Bangla and Hindi data often consists of custom, small or low-quality annotated data, which are limiting in flexibility of the model and performance.

These missing links in the currently existing corpus of knowledge prompted development and deployment of the present project: the multilingual and powerful system of AI Chatbots that will be able to execute real-life functions such as but not limited to Bangla and Hindi generation, translation, summarization, drug information processing, rental prediction, and mathematical calculations. We propose a blend of both classical models of ML, the transformer models of today, and an implementation to create conversational AI where a scalable and extendable model can be established to handle all sorts of linguistic scenarios.

III. METHODOLOGY

A. System Design

The architecture of the AI Chatbot system is based on the modular, controller-based architecture that allows flexibility, scalability and maintenance. The system contains several pre-trained AI/ML models and API services that can make multilingual NLP applications of language generation, translation,

summarization, mathematical computation and information retrieval.

The overall design can be represented by the following **block diagram**:

English NLP	Bangla NLP (T5,GPT)	Hindi NLP (Gemini API)
Math Solver (SymPy)	Drug Info (CSV Search)	Rental Search

In addition, **UML class diagrams** were used during planning to design interactions between core modules such as Chatbot, BanglaGenerator, Translator, DrugInfoProcessor, etc. All components follow the principles of **separation of concerns**, where each module is responsible for one specific functionality.

B. Hardware and/or Software Components

The AI Chatbot mechanism is purely computerized and there is no need to have the hardware elements involved. It is designed by means of Python libraries, ML/NLP models, and open-source tools. It is designed to focus on modular code, command-line interface (CLI), and external APIs integration.

Some of the key elements are:

- **Dataset:** Bangla and Hindi text custom and open-source datasets Input.
- **Preprocessing:** Text tokenization, stopword removal, and input normalization
- **Models:** T5 as summariser, GPT based Bangla generation, SymPy as math solver and Gemini API to Hindi response solver
- **Architecture:** Modular router architecture controller architecture
- **Interface:** Console program (command line)

TABLE I
EMPLOYED SOFTWARE TOOLS

Tool/Library	Function	Similar Tools	Why Selected
Python 3.10	Core development language	JavaScript, Java	High compatibility with ML libraries
Transformers (Hugging Face)	Pretrained NLP models	OpenNMT, Fairseq	Access to SOTA models & APIs
SymPy	Solving math expressions	NumPy, MathJS	Symbolic computation support
Pandas	CSV data handling	Excel, SQL	Easy manipulation of structured data
Gemini API	Hindi language generation	ChatGPT API	Optimized for multilingual responses
Git & GitHub	Version control, collaboration	GitLab, Bitbucket	Widely used, integrated CI/CD tools

C. Software Implementation

The implementation phase consisted of developing and integrating the core modules based on specific functionalities. Each module was first tested independently and later integrated under a unified controller system.

Steps followed during implementation:

- 1) **Module-wise development:** All the files like `bangla_generator.py`, `math_solver.py`, `translator.py`

were individually done so that modular testing could be achieved.

- 2) **Configuration/Model Loading:** The T5 and GPT models on Hugging Face were loaded with the pre-trained checkpoints on Bangla text summarization and generation. The Hindi text was transcribed through the Gemini API of Google.
- 3) **Function of Routings Logic:** The `chatbot.py` script is the controller and it dispatches the strings (CLI arguments) to the appropriate function/module.
- 4) **Text Preprocessing:** Cleaning of inputs was done by preprocessing functions which involved removing punctuations, lowering casing, and normalizing to have improved accuracy of inferences.
- 5) **Command-line Interface (CLI):** The last implementation was carried out in the form of a terminal-based command-line interface (CLI) tool and users would be able to send arguments as to which task they want to execute.
- 6) **Validation and testing:** All the modules were tested using variety of sample inputs to ascertain validity and integrity and were then brought to complete the application.

The software implementation adhered to clean coding practices, modularity, and reuse, allowing easy extension for new language or task modules in the future.

IV. INVESTIGATION/EXPERIMENT, RESULT, ANALYSIS AND DISCUSSION

A. Experiments and Evaluation Metrics

To test the usefulness of AI Chatbot system, a set of experiments was carried out over various modules such as the text generation, summarization, translation, drug information retrieval, rental prediction and mathematical computation.

The multiple inputs were used to test each module with the aim to measure

- Accuracy (in case of structured activities such as solutions of mathematical problems and search of drugs),
- Fluency and Coherence (used in generative work such as a summary and language generation),
- Latency / Response Time (seconds).
- User Satisfaction Score (based on reviews of 10 test users 1-5 scale).

B. Results and Figures

C. Analysis

The modules in the chatbot were first tested individually and afterward, in a unit. The multilingual generation modules (Bangla and Hindi) were producing decent results, but Bangla had a slight delay in inference time since it was loading localized models. The translation part and the summarization part were found to perform well both in terms of quality (measured through human evaluation) and quantitatively (measured through BLEU and ROUGE scores).

The drug info module gave the very accurate and rapid response whose only shortcoming was when improperly spelled drug names were used in the query. The math solver had an impeccable reading of symbols and solving even in nested expressions.

The rental prediction module showed high accuracy and consistency with user-provided data during testing. The use of basic regression allowed fast execution with decent generalization. The relatively low Mean Absolute Error indicates good performance even in sparse data conditions.

D. Discussion

The creation of the Chatbot that relied on AI and Machine Learning technologies was successful in integrating various capabilities of natural language processing into one modular system. As it included the modules of multilingual generation (Bangla, Hindi, and English), translation, summarization, mathematical computation, drug information retrieval, and rental price prediction, the system proved to be versatile and adaptable against a broad variety of application scenarios. According to test results, structured task execution like mathematical problem solving and queries of drug databases were accurate and low-latency, and the math solver was faultless even with complex calculative expression. Generative tasks like summarization and translation resulted in coherent and context-sensitive output but showed some delay in processing large input, or loading language models localized to a region. We can run according to the attributes. That means what we wanted to do, we can directly chose in the command section.

One exceptional strength that was offered by the system was the modular architecture which facilitated independent testing, simplified the maintenance process and even possible expansion in capacity to accommodate new languages or areas. Bangla and Hindi additions have garnered easier use of the language through multilingual areas such as South Asia to encourage digital inclusiveness among non-English citizens. Nevertheless, there are still some restraints. The spelling of the medicine search engine was not tolerant of spelling differences making the facility to be less efficient to its users who made minor typing mistakes. The modules of predicting rent and retrieving medicines based on based on relatively small, tailored datasets could influence the accuracy in diverse conditions closer to reality.

On the whole, the architecture, functionality, and performance of the chatbot ensure that it can be deployed into the

real world, especially in a learning, healthcare, and information services environment. It has low pro-processing needs that can be locally deployed without using high-end equipment, which is in line with accessibility and sustainability objectives. Fine-tuning changes—including the incorporation of fuzzy match queries, the increase of datasets, the adoption of a mobile interface and the promotion of real-time-learning characteristics—would also increase the usefulness, expandability and stability of the system in busy user environments.

V. IMPACTS OF THE PROJECT

A. Impact on Societal, Health, Safety, Legal, and Cultural Issues

The AI Chatbot project has significant implications in the global value of the society and culture with improved access to digital distribution of services accessible through Bangla, Hindi and English. Being multilingual guarantees the fact that people who are not fluent in English can also use intelligent systems and be digital-inclusive, decreasing the digital gap in South Asia.

Health-wise, the chatbot has a drug information service that helps customers access important information about drugs, the side effects, and tips on using a medicine - this gives a person power to make decisions about his/her health. It should not replace medical advice, but it is a part of health literacy in areas where healthcare professionals are not accessible enough.

As far as legal and ethical responsibility is concerned, the system is made transparent and is educational rather than automated or the collection of sensitive data. Nevertheless, there should also be caution when the given tools are utilized publicly so that the privacy of data would be guaranteed, no false information would be disseminated, and biases would be minimized, especially with response generation and prediction modules.

This chatbot is culturally sensitive to language diversity and supports the sentences of underrepresented languages, such as Bangla and Hindi, which help promote the local identity and make AI technologies used in that region more widely accepted.

B. Impact of this project on environment and sustainability

Being an exclusively software-based initiative, the AI Chatbot does not produce direct effects to the physical surroundings, in this aspect, at least, leaving less significant footprint than hardware-based systems. Nevertheless, it helps to achieve sustainability in a number of indirect but significant ways:

- 1) Having the user get the assistance they need (drug instructions, rental pricing estimates, language translation, etc.) online makes it less necessary to require printed copies and on-site counseling, resulting in the reduced usage of the resources.
- 2) The system is energy efficient and scalable, with the possibility of implementing low-power computing infrastructure with no special hardware.

- 3) Its reuse (through its modular architecture) and open source flexibility enhance its sustainability with respect to the engineering of software.

Although the algorithm itself does not report and resolve the explicit environment-related problems, it preconditions the subsequent extension of the chatbot in the area of environmental education, eco-resources management, or environmentally friendly decision-making.

VI. PROJECT PLANNING AND BUDGET

Effective planning was critical for the successful execution of this chatbot project. A Gantt chart was used to plan and monitor progress across key milestones, from the literature review to final testing and documentation. The planning was divided into several phases over 12 weeks.

TABLE II
PROJECT TIMELINE

Task	Week 1–2 May 21–28	Week 3–4 June 4–18	Week 5–6 July 2–9	Week 7–8 July 16–23	Week 9–10 July 23–30	Week 11–12 Aug 6–13
Literature Review & Problem Analysis	X	X				
Dataset Collection & Preprocessing		X	X			
Model Selection & Setup			X	X		
Module-Wise Implementation				X	X	
Integration & CLI Development					X	
Testing & Evaluation					X	X
Documentation & Final Report						X

As a software-centric project, the budget focused mainly on software tools, cloud services, and optional hardware for future expansion. Most tools used were open-source, significantly reducing the cost.

TABLE III
BUDGET TABLE

Item / Tool	Purpose	Estimated Cost (BDT)
Python (Open Source)	Core programming language	0
Hugging Face Transformers	Pre-trained NLP models	0
SymPy	Symbolic math solving	0
Gemini API	Hindi text generation	0
Google Colab (optional)	Cloud-based training/testing	0
Custom Dataset Collection	Manual and online data gathering	0
Internet and Software Utilities	Development and testing support	0
GitHub (Free Tier)	Version control	0
Documentation/Printing	Report preparation	500
Total Estimated Cost (BDT)		500

VII. COMPLEX ENGINEERING PROBLEMS AND ACTIVITIES

A. Complex Engineering Problems (CEP)

The AI and Machine Learning-Based Chatbot addresses multiple **Complex Engineering Problems (CEP)**, as outlined below:

TABLE IV
COMPLEX ENGINEERING PROBLEM ATTRIBUTES

Attributes		Addressing the complex engineering problems (P) in the project
P1	Depth of knowledge required (K3–K8)	This project demands knowledge in various domains, including Natural Language Processing (K3), Python programming and ML frameworks (K4), software architecture design and API integration (K5), use of engineering tools like Hugging Face, SymPy, and Google Gemini API (K6), awareness of AI biases and ethical issues (K7), and interpretation of scientific research and evaluation metrics (K8).
P2	Range of conflicting requirements	Trade-offs were required between response speed and model complexity. For example, transformer-based models provide high accuracy but increase latency. Choosing between general-purpose models and domain-specific performance also required balancing.
P3	Depth of analysis required	Designing a unified chatbot involved selecting between multiple language models (T5, GPT, mBERT), CLI architectures, and routing logic. Evaluation of alternatives was essential for determining scalability, latency, and multilingual performance.
P4	Familiarity of issues	Working with Bangla and Hindi NLP presented unique challenges due to a lack of large-scale pre-trained models. Custom datasets and language-specific preprocessing techniques had to be researched and applied, which are less familiar in mainstream AI development.

B. Complex Engineering Activities (CEA)

The development of this chatbot involved various **Complex Engineering Activities (CEA)** in line with global engineering practices, user needs, and technological integration.

TABLE V
COMPLEX ENGINEERING ACTIVITIES ATTRIBUTES

Attributes		Addressing the complex engineering activities (A) in the project
A1	Range of resources	The project involved a variety of resources including human expertise in NLP and software engineering, computational tools (Python, SymPy, Hugging Face Transformers), online APIs (Gemini), custom datasets for Rental and Medicine suggestion, and cloud platforms like Google Colab. While hardware was not directly involved, software dependencies and data infrastructure were key technical resources.
A2	Level of interactions	The chatbot's design involved team collaboration across modules, as well as consideration of end-user interaction with multiple languages. Stakeholder input, such as from users needing Bangla/Hindi support, rental information, and healthcare assistance, influenced feature prioritization and interface design.
A3	Innovation	This project introduced an innovative fusion of multiple AI functionalities—summarization, translation, math solving, drug search—within a single, modular CLI-based chatbot system. Its multilingual support for Bangla and Hindi sets it apart from conventional English-only systems.

VIII. JUSTIFICATION OF COMPLEX ENGINEERING PROBLEM ATTRIBUTES

A. Why P1: Depth of Knowledge Required (K3–K8) is necessary for our project?

The inclusion of **P1: Depth of Knowledge Required (K3–K8)** is essential in this project because the development of an AI and Machine Learning-Based Chatbot involves a wide range of advanced engineering concepts. It requires a strong foundation in programming logic, algorithms, and mathematics (K3), along with discipline-specific expertise in natural language processing, machine learning models, and symbolic computation (K4).

The design of a modular, controller-based architecture capable of integrating multiple pre-trained models such as T5, GPT, and SymPy reflects a need for comprehensive system design and modeling skills (K5). Furthermore, the project utilizes modern engineering tools and platforms including Hugging Face Transformers, Gemini API, and version control via Git (K6), all of which are integral to contemporary AI development.

The chatbot also addresses broader engineering concerns such as ethical AI use, social impact, and multilingual accessibility (K7), demonstrating the importance of responsible technology design. Finally, the project required reviewing cutting-edge research, identifying existing limitations, and developing a novel, scalable solution (K8), all of which indicate a high level of academic and engineering competence.

Therefore, the depth of knowledge encapsulated in K3 to K8 is not only relevant but critical to the success and impact of this project.

IX. CONCLUSIONS

A. Summary

This project presents the development of an intelligent, multilingual chatbot system leveraging Artificial Intelligence (AI) and Machine Learning (ML) techniques. The chatbot is capable of handling multiple natural language processing (NLP) tasks such as question-answering, translation, summarization, mathematical computation, and domain-specific information retrieval (e.g., medicine lookup and rental price prediction). The system was built using Python and integrates state-of-the-art models like T5, GPT, SymPy, and the Gemini API, all organized within a modular, CLI-based framework.

The primary contribution of this work lies in its ability to combine multiple capabilities into a unified, scalable architecture while supporting underrepresented languages such as Bangla and Hindi. Extensive testing across individual modules has shown promising accuracy, responsiveness, and relevance, making it a practical assistant in real-world applications such as education, healthcare, and smart search tools.

B. Limitations

Despite the successful implementation of core functionalities, the project has several limitations:

- **Limited Training Data:** Medicine and Rental modules relied on limited or custom datasets, which can reduce accuracy in real-world diverse scenarios.
- **No Mobile Interface:** The system is built for local CLI use and does not support mobile interfaces yet.
- **Model Performance Constraints:** Some tasks, especially translation and summarization, showed latency or quality issues depending on the size of the input and model complexity.
- **Spelling Sensitivity:** The medicine lookup module is sensitive to misspellings or variations in drug names.
- **No Real-time Adaptation:** Although modular, the current architecture does not support continuous learning from user input or adaptive feedback loops.

C. Future Improvement

Several enhancements can be made in future iterations of this project:

- **Mobile Interface:** Deploy the chatbot on a mobile platform for wider accessibility and usability.
- **Enhanced Dataset and Fine-Tuning:** Incorporate large, high-quality medicine and Rental datasets to fine-tune language models for better fluency and accuracy.
- **Spelling Correction and Fuzzy Matching:** Integrate spell-checking algorithms or fuzzy string matching to make the system more tolerant to user input errors.
- **Multimodal Capabilities:** Extend the system to support voice input/output or image-based question answering for improved interaction.
- **Real-time Learning:** Add reinforcement learning mechanisms to improve responses over time based on user feedback.
- **Privacy and Security:** Implement user data protection features and ethical guidelines, especially if used in health-related contexts.

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